SofiMind Data Consulting

Independent Clinical Data & Outcomes Analysis

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Why Data Analytics Matters in Clinical Studies

- New drugs/medicine development
- Identifies patterns in outcomes, risk factors and groups
- Improves data quality through cleaning, validation and transparency
- Supports better decisions

Transforms raw clinical data into actionable insights

Case Studies

- Two different datasets
- Cancer accounts for 15% of deaths in the world (2021)
- More than 6500 new clinical studies only in 2021
- Most focusing on treatments and survival

01. Oral cancer and survival

02. Lung cancer and survival

Does **treatment** type influence the **survival** of those cancer patients?

Case Study 1:

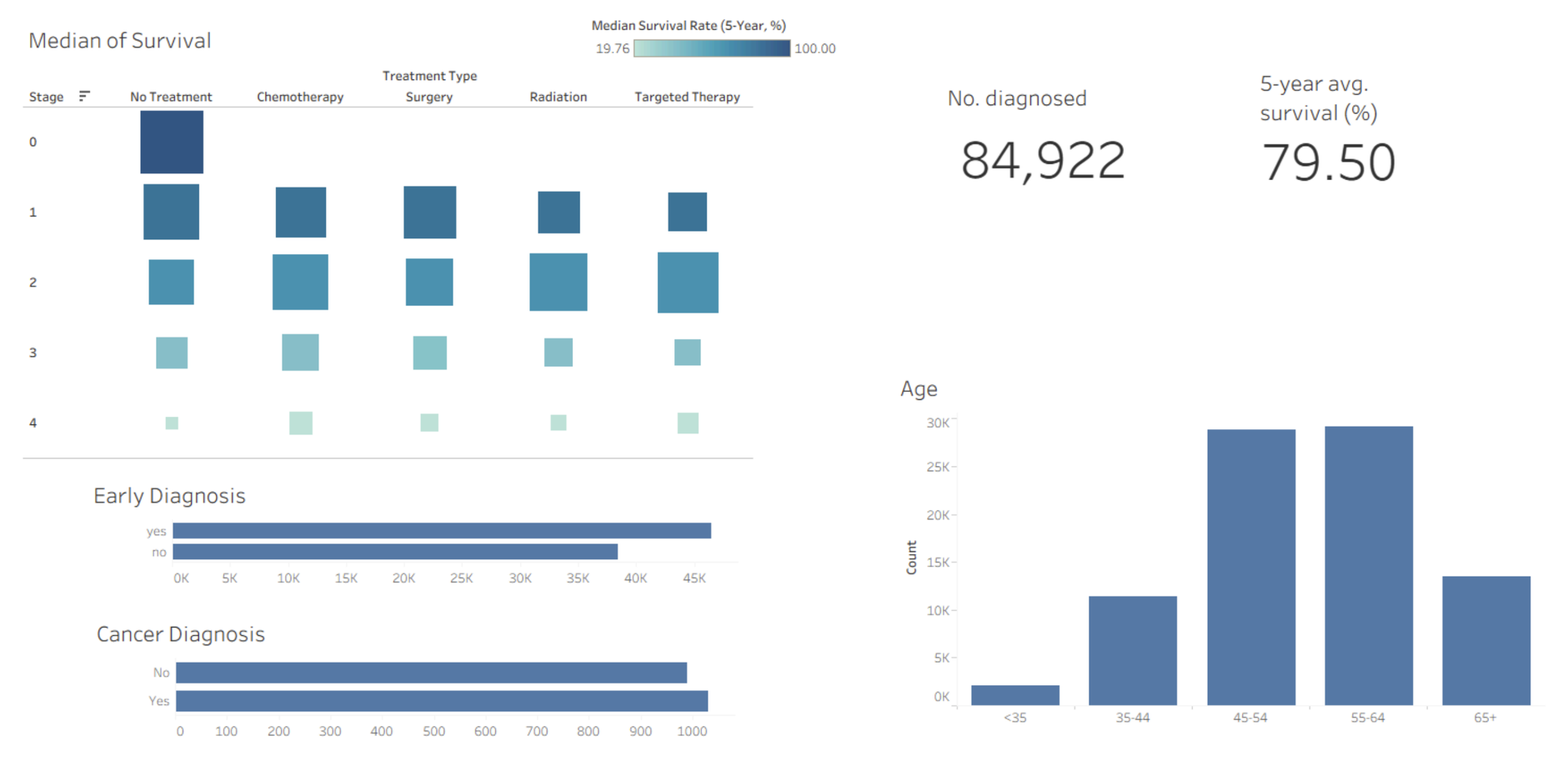
- Oral cancer: mouth and the back of the throat
- Risk factors: **genetics, lifestyle factors and viral infections**
- Symptoms: white/red patches, lumps, difficulty swallowing, pain, bleeding
- Different treatment types

Dataset and Tools

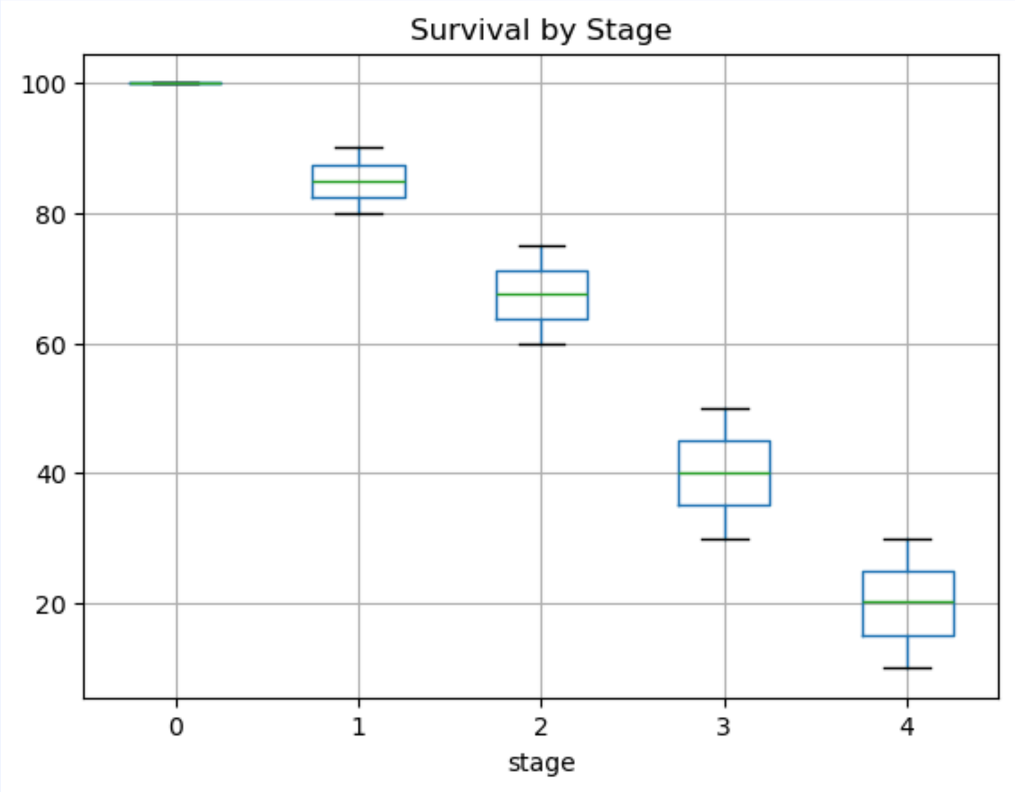
- Source: Kaggle
- Oral Cancer Prediction Dataset

- Python libraries:
  - pandas (data cleaning/manipulation)
  - matplotlib, seaborn (visualization)
  - scipy (statistics)
- SQL
- Tableau

Results/Demo



Results/Demo

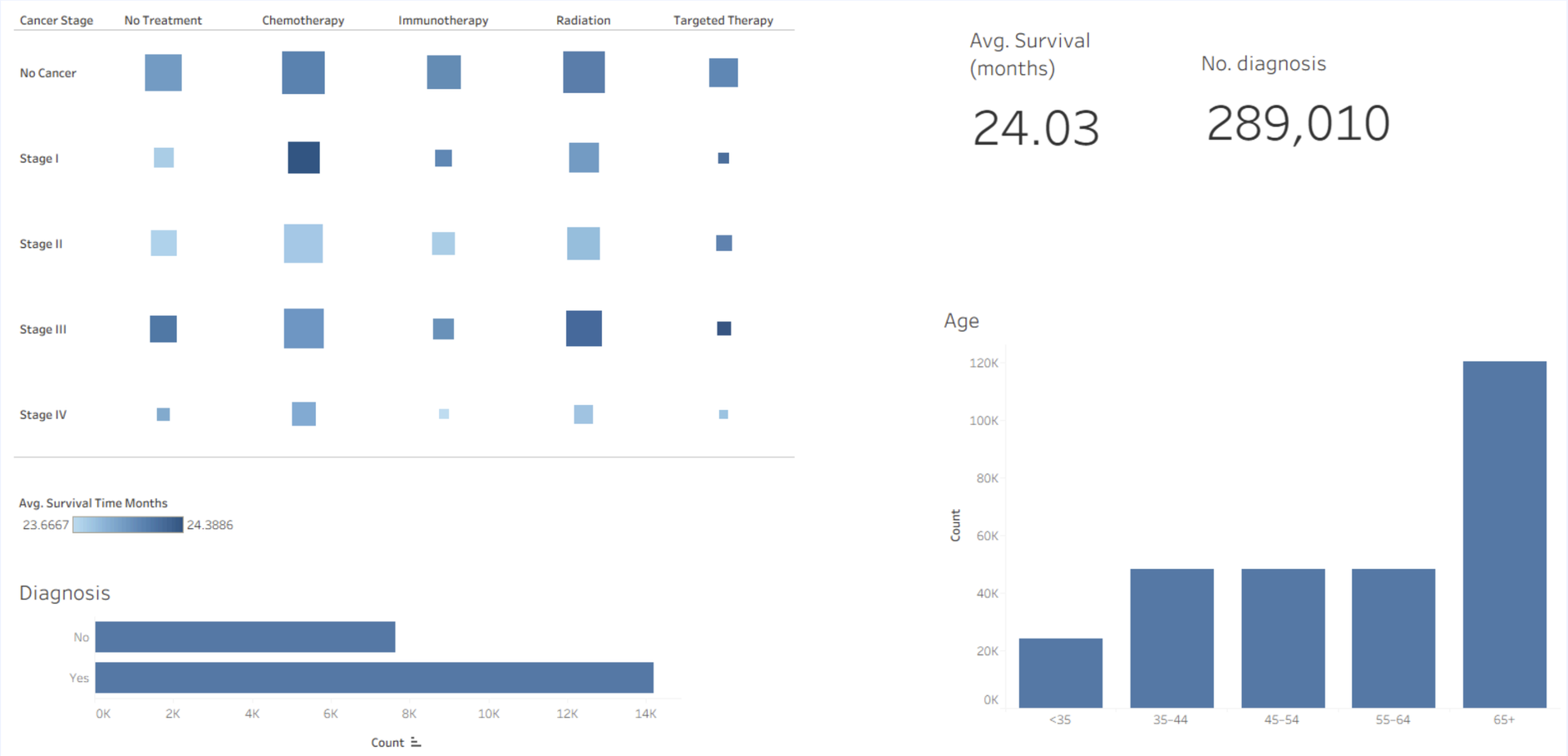


$p < 0.01$

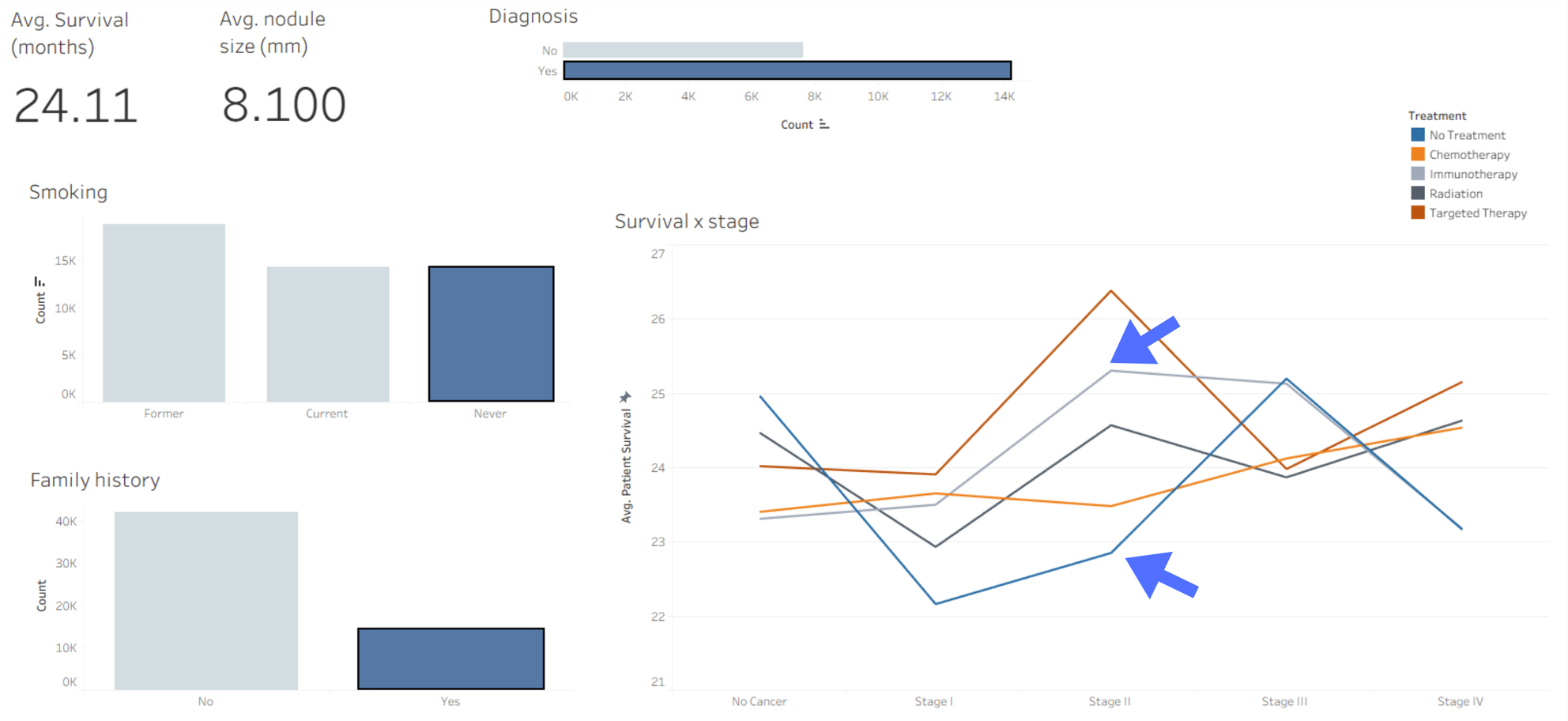
Case Study 2

- Lung cancer
- Risk factors: **smoking, second-hand smoke, genetics, dangerous substances (like asbestos)**
- Symptoms: persistent cough, chest pain, shortness of breath, bleeding
- Different treatment types

Results/Demo



Results/Demo



p = 0.04

Conclusions

Statistical difference (expected) between stages of the disease for survival.

No significant survival differences found comparing treatments in oral cancers or in lung cancer datasets.

For lung cancer: patients who never smoked and had family history have increased survival when treated with immuno-therapy.

High treatment cost and more days lost on patients treated with targeted therapy.

Disclaimer: These results are based only on these datasets and are exploratory. They do not prove real-world clinical effects or causality.

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Thank you very much!

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